

MAKE CONNECTIONS

Genomics, Cell Signaling, and Cancer

Modern medicine that melds genome-wide molecular studies with cell-signaling research is transforming the treatment of many diseases, such as breast cancer. Using microarray analysis (see Figure 20.13) and other techniques, researchers measured the relative levels of mRNA transcripts for every gene in hundreds of breast cancer tumor samples. They identified four major subtypes of breast cancer that differ in their expression of three signal receptors involved in regulating cell growth and division:

- Estrogen receptor alpha ($ER\alpha$)
- Progesterone receptor (PR)
- HER2, a type of receptor called a receptor tyrosine kinase (RTK); other RTKs are also present, such as HER3

($ER\alpha$ and PR are steroid receptors; see Figure 11.9.) The absence or excess expression of these receptors can cause aberrant cell signaling, leading in some cases to inappropriate cell division, which can contribute to cancer (see Figure 18.24).



▲ A research scientist examines DNA-sequencing data from breast cancer samples.

